

MongoDB Lab - Day 3 (CRUD Operations)

Name - Ankita Das

Roll No - CSE CSF 017

Schema:

{ roll no, name, age, department, city, marks }

Initial Data:

Roll No	Name	Age	Department	City	Marks
1	Amit	20	CSE	Kolkata	78
2	Neha	19	IT	Delhi	65
3	Ravi	21	CSE	Mumbai	82
4	Pooja	22	ECE	Chennai	70
5	Karan	20	IT	Kolkata	55

Database Name: collegeDB_'CSF017'

Collection Name: students_CSE017

Lab Questions:

Q1. Display all student records from the collection.

Query: `db.students_CSF017.find()`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2621'),
  roll_no: 1,
  name: 'Amit',
  age: 20,
  department: 'CSE',
  city: 'Kolkata',
```

```
marks: 78
},
{
  _id: ObjectId('6980439596cb41c9791e2622'),
  roll_no: 2,
  name: 'Neha',
  age: 19,
  department: 'IT',
  city: 'Delhi',
  marks: 65
},
{
  _id: ObjectId('6980439596cb41c9791e2623'),
  roll_no: 3,
  name: 'Ravi',
  age: 21,
  department: 'CSE',
  city: 'Mumbai',
  marks: 82
},
{
  _id: ObjectId('6980439596cb41c9791e2624'),
  roll_no: 4,
  name: 'Pooja',
  age: 22,
  department: 'ECE',
  city: 'Chennai',
  marks: 70
},
{
  _id: ObjectId('6980439596cb41c9791e2625'),
```

```
roll_no: 5,  
name: 'Karan',  
age: 20,  
department: 'IT',  
city: 'Kolkata',  
marks: 55  
}  
]
```

Q2. Display only the name and department of all students.

```
Query: db.students_CSF017.find(  
... {},  
... { name: 1, department: 1, _id: 0 }  
... )
```

Output: ...

```
[  
  { name: 'Amit', department: 'CSE' },  
  { name: 'Neha', department: 'IT' },  
  { name: 'Ravi', department: 'CSE' },  
  { name: 'Pooja', department: 'ECE' },  
  { name: 'Karan', department: 'IT' }  
]
```

Q3. Find students belonging to the CSE department.

```
Query: db.students_CSF017.find({ department: "CSE" })
```

Output: [

```
{  
  _id: ObjectId('6980439596cb41c9791e2621'),  
  roll_no: 1,  
  name: 'Amit',
```

```
    age: 20,
    department: 'CSE',
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('6980439596cb41c9791e2623'),
    roll_no: 3,
    name: 'Ravi',
    age: 21,
    department: 'CSE',
    city: 'Mumbai',
    marks: 82
  }
]
```

Q4. Find students whose age is greater than 20.

Query: `db.students_CSF017.find({ age: { $gt: 20 } })`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2623'),
  roll_no: 3,
  name: 'Ravi',
  age: 21,
  department: 'CSE',
  city: 'Mumbai',
  marks: 82
},
{
  _id: ObjectId('6980439596cb41c9791e2624'),
  roll_no: 4,
```

```
name: 'Pooja',
age: 22,
department: 'ECE',
city: 'Chennai',
marks: 70
}
]
```

Q5. Find students who scored less than 70 marks.

Query: `db.students_CSF017.find({ marks: { $lt: 70 } })`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2622'),
  roll_no: 2,
  name: 'Neha',
  age: 19,
  department: 'IT',
  city: 'Delhi',
  marks: 65
},
{
  _id: ObjectId('6980439596cb41c9791e2625'),
  roll_no: 5,
  name: 'Karan',
  age: 20,
  department: 'IT',
  city: 'Kolkata',
  marks: 55
}
]
```

Q6. Display students belonging to Kolkata city.

Query: `db.students_CSF017.find({ city: "Kolkata" })`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2621'),
  roll_no: 1,
  name: 'Amit',
  age: 20,
  department: 'CSE',
  city: 'Kolkata',
  marks: 78
},
{
  _id: ObjectId('6980439596cb41c9791e2625'),
  roll_no: 5,
  name: 'Karan',
  age: 20,
  department: 'IT',
  city: 'Kolkata',
  marks: 55
}
]
```

Q7. Sort students by age in ascending order.

Query: `db.students_CSF017.find().sort({ age: 1 })`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2622'),
  roll_no: 2,
  name: 'Neha',
  age: 19,
```

```
department: 'IT',
city: 'Delhi',
marks: 65
},
{
  _id: ObjectId('6980439596cb41c9791e2621'),
  roll_no: 1,
  name: 'Amit',
  age: 20,
  department: 'CSE',
  city: 'Kolkata',
  marks: 78
},
{
  _id: ObjectId('6980439596cb41c9791e2625'),
  roll_no: 5,
  name: 'Karan',
  age: 20,
  department: 'IT',
  city: 'Kolkata',
  marks: 55
},
{
  _id: ObjectId('6980439596cb41c9791e2623'),
  roll_no: 3,
  name: 'Ravi',
  age: 21,
  department: 'CSE',
  city: 'Mumbai',
  marks: 82
},
```

```
{
  _id: ObjectId('6980439596cb41c9791e2624'),
  roll_no: 4,
  name: 'Pooja',
  age: 22,
  department: 'ECE',
  city: 'Chennai',
  marks: 70
}
]
```

Q8. Sort students by marks in descending order.

Query: db.students_ CSF017.find().sort({ marks: -1 })

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2623'),
  roll_no: 3,
  name: 'Ravi',
  age: 21,
  department: 'CSE',
  city: 'Mumbai',
  marks: 82
},
{
  _id: ObjectId('6980439596cb41c9791e2621'),
  roll_no: 1,
  name: 'Amit',
  age: 20,
  department: 'CSE',
  city: 'Kolkata',
  marks: 78
}
```



```
},
{
  _id: ObjectId('6980439596cb41c9791e2624'),
  roll_no: 4,
  name: 'Pooja',
  age: 22,
  department: 'ECE',
  city: 'Chennai',
  marks: 70
},
{
  _id: ObjectId('6980439596cb41c9791e2622'),
  roll_no: 2,
  name: 'Neha',
  age: 19,
  department: 'IT',
  city: 'Delhi',
  marks: 65
},
{
  _id: ObjectId('6980439596cb41c9791e2625'),
  roll_no: 5,
  name: 'Karan',
  age: 20,
  department: 'IT',
  city: 'Kolkata',
  marks: 55
}
]
```

QG. Display only the first three student records.

Query: `db.students_ CSF017.find().limit(3)`

Output: [

```
{
  _id: ObjectId('6980439596cb41c9791e2621'),
  roll_no: 1,
  name: 'Amit',
  age: 20,
  department: 'CSE',
  city: 'Kolkata',
  marks: 78
},
{
  _id: ObjectId('6980439596cb41c9791e2622'),
  roll_no: 2,
  name: 'Neha',
  age: 19,
  department: 'IT',
  city: 'Delhi',
  marks: 65
},
{
  _id: ObjectId('6980439596cb41c9791e2623'),
  roll_no: 3,
  name: 'Ravi',
  age: 21,
  department: 'CSE',
  city: 'Mumbai',
  marks: 82
}
]
```

Q10. Update the marks of student Neha to 72.

```
Query: db.students_ CSF017.updateOne(  
... { name: "Neha" },  
... { $set: { marks: 72 } }  
... )
```

```
Output: {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

Q11. Increase the marks of all students by 5.

```
Query: db.students_ CSF017.updateMany(  
... {},  
... { $inc: { marks: 5 } }  
... )
```

```
Output: {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 5,  
  modifiedCount: 5,  
  upsertedCount: 0  
}
```

Q12. Delete the student having roll number 5.

```
Query: db.students_CSE326.deleteOne({ roll_no: 5 })  
Output: { acknowledged: true, deletedCount: 1 }
```

Q13. Find students whose age is between 20 and 22.

Query: db.students_CSF017.find({

... age: { \$gte: 20, \$lte: 22 }

... })

Output: [

{

_id: ObjectId('6980439596cb41c9791e2621'),

roll_no: 1,

name: 'Amit',

age: 20,

department: 'CSE',

city: 'Kolkata',

marks: 83

},

{

_id: ObjectId('6980439596cb41c9791e2623'),

roll_no: 3,

name: 'Ravi',

age: 21,

department: 'CSE',

city: 'Mumbai',

marks: 87

},

{

_id: ObjectId('6980439596cb41c9791e2624'),

roll_no: 4,

name: 'Pooja',

age: 22,

department: 'ECE',

city: 'Chennai',

marks: 75

```
}  
]
```

Q14. Display students belonging to either CSE or IT department.

Query: `db.students CSF017.find({
... department: { $in: ["CSE", "IT"] }
... })`

Output: [

```
{  
  _id: ObjectId('6980439596cb41c9791e2621'),  
  roll_no: 1,  
  name: 'Amit',  
  age: 20,  
  department: 'CSE',  
  city: 'Kolkata',  
  marks: 83  
},  
{  
  _id: ObjectId('6980439596cb41c9791e2622'),  
  roll_no: 2,  
  name: 'Neha',  
  age: 19,  
  department: 'IT',  
  city: 'Delhi',  
  marks: 77  
},  
{  
  _id: ObjectId('6980439596cb41c9791e2623'),  
  roll_no: 3,  
  name: 'Ravi',  
  age: 21,
```

```
department: 'CSE',  
city: 'Mumbai',  
marks: 87  
}  
]
```

```
_id: ObjectId('6980439596cb41c9791e2621')  
roll_no : 1  
name : "Amit"  
age : 20  
department : "CSE"  
city : "Kolkata"  
marks : 83
```

```
) _id: ObjectId('6980439596cb41c9791e2622')  
roll_no : 2  
name : "Neha"  
age : 19  
department : "IT"  
city : "Delhi"  
marks : 77
```

```
_id: ObjectId('6980439596cb41c9791e2623')  
roll_no : 3  
name : "Ravi"  
age : 21  
department : "CSE"  
city : "Mumbai"  
marks : 87
```

```
_id: ObjectId('6980439596cb41c9791e2624')  
roll_no : 4  
name : "Pooja"  
age : 22  
department : "ECE"  
city : "Chennai"  
marks : 75
```